

Automating the obstacle course

with Flowproof 0.12.2

Human intent at record time. Deterministic proof at replay.

40

pages

28

passing

12

refused

- Verified against the 0.12.2 authoring contract

The first half teaches the mechanism.

The second half lets every obstacle name the technique it needs.

Human intent

Write the action in ordinary words.

Grounded recording

The model chooses only from the live inventory.

Deterministic replay

Selectors live in the trace, not in the author's prose.

Honest refusal

A false green is worse than a loud limitation.

Forty exercises

Twenty-eight pass; twelve document a real boundary.

Record once. Replay from then on.

The model participates only while the human intent is grounded.

Record

Live app + human steps
produces a validated trace

Replay

Trace only
zero authoring-model calls

```
flowproof record order.flow.yaml
flowproof run    order.flow.yaml
```

```
name: Order status
app: web
url: https://example.com/orders
steps:
  - Enter ABC in the order ID field
  - Search for the order
  - assert: page shows Shipped
```

Describe the thing you mean.

Different wording can express the same intent because the live scene supplies the possible targets.

Save the form

the visible Save control

Could you save this?

the same intent, conversationally

Use the second Delete button

one of several look-alikes

Choose Grace Hopper's row

a target identified by nearby text

Put Tosca in the embedded form

a field inside a same-origin frame

Relationships make selectors unnecessary.

A human names the value by what surrounds it; Flowproof records the deterministic scoped target.

steps:

- Remember the value beside "order id" as the order ID
- Enter the remembered order ID in the offer ID field
- assert: page shows You solved this automation problem

- The author names the relationship, not the DOM structure

Say what the person does.

Plain language covers direct, relational, positional, multi-control, frame, and keyboard actions.

Save the form

press a control

Click the right half of the button

click a meaningful point

Choose A, B and C together

commit several options

Bring the footer into view

scroll to an off-screen target

Drag Row 1 into Done

press, move while held, release

Move focus to the next field

press Tab and commit the field

A drag must be followed by proof.

The drop effect belongs to the application, so the next step must assert what changed.

steps:

- Drag task 1 into the todo drop area
- assert: the "css:tbody.droparea" shows Write recipe

- **Actions stay human; this assertion is explicit because the drop area has no usable name**

Assertions wait for outcomes, not time.

A fixed sleep guesses. An assertion waits until its condition is true or its bound expires.

- assert: page shows Saved
- assert: the "SEND" is enabled within 40s
- assert: the "Total" shows 42
- assert: the "Basket" is not visible
- assert: the "css:.row" appears 5 times

When the app invents a value, remember it.

The value is read fresh during record and replay; only the human name appears in the specification.

steps:

- Remember the offer ID as the offer
- Enter the remembered offer in Reference
- Remember the number of displayed rows as the row count
- Enter the remembered row count in Count

- **No generated value is frozen into the trace**

Pin what drifts, then read the page.

A seed, clock, and viewport make the environment reproducible; they do not replace observation.

```
browser:
  viewport: { width: 1280, height: 900 }
  random:
    seed: 1
  clock:
    at: "2026-01-15T12:00:00Z"
    timezone: "Europe/Berlin"
```

- 1 Pin the environment.
- 2 Run with an intentionally wrong assertion.
- 3 Read the real value and update the flow.

Read values off the page. Do not predict them.

Other scripts may consume random draws first. The seed fixes the sequence, not the position.

The stable test is not “our PRNG simulation agrees.”

The stable test is “the page displayed this value, the flow used it, and replay proved the same relationship again.”

- A pinned viewport also prevents controls from drifting under occluders

Repeat and when remain explicit.

Control structure is deterministic; the actions nested inside it remain ordinary human intent.

- repeat:
 - until: page shows Enough
 - max: 15
 - steps:
 - Press the counter button again
- when: the recovery button is not visible
 - steps:
 - Enable recovery mode

Conditions read state and never wait.

Keep them narrow enough to distinguish the intended state from similar words elsewhere on the page.

page shows X

the visible surface contains X

the target shows X

one resolved element contains X

the target is visible

it resolves and is rendered

A is greater than B

both values parse as numbers

- **A broad page condition can match an unrelated heading**

The frame is part of the target.

People describe the embedded surface; the recording stores an ordinary deterministic framed target.

steps:

- Scroll the embedded challenge to 147 pixels
- Enter Tosca in the text field inside the embedded challenge
- Submit the answer

Flowproof refuses shortcuts a person could not take.

The framework treats a trustworthy failure as more valuable than an unearned green.

No invented selectors

The model must choose a listed live target.

No click through an occluder

The real hit target must receive the click.

No hidden human target

A rendered inventory cannot name an invisible control.

No synthetic success

An app-defined effect must be asserted.

A false green is worse than a refusal.

Obstacle 51130 reports success even though its popup never opens. Recording that pass would prove nothing.

REFUSE

A passing report must correspond to working software.

Human intent is default. Structure stays explicit.

The source remains readable without making replay probabilistic.

Plain scalar actions

- Any natural wording
- Grounded against the live scene
- Validated target inventory
- Deterministic trace output

Structured proof

- assert: expected outcome
- repeat: condition and bound
- when: deterministic branch
- Explicit failure semantics

Every exercise names its technique.

Green entries have a passing record and replay. Red entries document a real boundary.

12952

Twins

No completion path

14090

Dropdown table

Select several controls

16384

Test data in a service

Deadline shorter than the flow

19875

Tomorrow

Pin the clock

21269

Future Christmas

Pin the clock

22505

Ids are not everything

No completion path

23292

Todo list

Drag and prove the drop

24499

And counting

Pin randomness

30034

Red stripe

Ground a small visual target

32403

Math

Read, then answer

Obstacle map, continued.

Green entries have a passing record and replay. Red entries document a real boundary.

33678 **Wait a moment**
Auto-wait an assertion

41036 **Table search**
Pin randomness

41038 **Halfway**
Click a meaningful point

41041 **Escape**
Literal text

51130 **Popup windows**
Vacuous green refused

41032 **Lots of rows**
Remember a count

41037 **Meeting scheduler**
Read a table relationship

41040 **Click me if you can**
Pointer cannot land

45618 **Tough cookie**
Focus and key commit

57683 **Confusing dates**
Pin randomness

Obstacle map, continued.

Green entries have a passing record and replay. Red entries document a real boundary.

60469 **Flying element**

No completion path

66666 **Hidden element**

Not human-groundable

70310 **The last row**

Remember the final cell

72946 **Get the number**

Read a fixed value

73588 **The obvious**

Select a generated value

64161 **Not a table**

Scoped live value

66667 **Empty**

Ground maze checkpoints

70924 **Errors occur**

Repeat and recover

72954 **Two times**

Stable visible text

73589 **Bubble sort**

Repeat and compare

Obstacle map, continued.

Green entries have a passing record and replay. Red entries document a real boundary.

73590

Find and fill

No completion path

78264

Addition

Pin randomness

81121

Again and again

Repeat to visible state

87912

Be fast

Deadline shorter than the flow

94441

Testing methods

Choose several options

73591

Find the changed cell

No completion path

81012

Extracting text

Read, then answer

82018

Reaction game

Frame-rate timing

92248

Fun with tables

No completion path

99999

Scroll into view

Work inside a frame

Twenty-eight passing solutions.

Twelve honest refusals.

Zero sketches.

Every green corresponds to a passing record and a passing replay.